# Single-Electron Field Identity (SEFI): A Geometric Resonance Ontology for the Electron

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(Dated: August 21, 2026)

The standard model of particle physics treats electrons as distinct point-like particles that happen to share identical properties: charge, mass, spin, and coupling behavior. Quantum field theory refines this picture by introducing a fermionic field whose excitations correspond to electrons, but the ontology remains effectively "many electrons" with identical excitations. In this work I propose a different starting point: all observable electrons are manifestations of a single underlying entity, expressed as repeated instances of one stable geometric resonance mode.

The Single-Electron Field Identity (SEFI) framework replaces the usual particle ontology with a curvature-based resonance ontology. The electron is modeled as a global mode of a warp-metric geometry, where charge, mass, spin, and stability arise from the structure of the mode itself rather than from separate fields or symmetry mechanisms. Nonlocality and coherence are then natural consequences of unified identity: if all electrons are the same entity, there is no need to invoke wavefunction collapse, hidden variables, many worlds, or retrocausality to explain correlated behavior.

I present a formal geometric construction in which the electron corresponds to the first stable harmonic of a curvature-resonance field, defined on a background metric with controlled warp and torsion. The SEFI model shows how electron properties emerge from the same underlying mode, unifying charge, mass, spin, and curvature in a single structure. The framework is supported by a set of warp-metric diagrams and visual models that map directly onto the equations, providing an unusually transparent bridge between geometric intuition and formal mathematics.

This paper outlines the core SEFI ontology, derives the basic electron mode structure, and discusses implications for nonlocality, identity, and the interpretation of quantum mechanics. The goal is not to replace existing calculational machinery, but to provide a coherent geometric foundation that explains why electrons behave as they do and why their properties are so rigidly uniform.

## I. INTRODUCTION

In conventional quantum theory, the electron is treated as a point-like particle with fixed intrinsic properties: charge $-e$, mass $m_e$, spin $1/2$, and a set of well-defined couplings. Quantum field theory refines this picture by introducing a spinor field $\psi(x)$ whose excitations correspond to electrons, but the ontology remains effectively multiplicity: there are many electrons, all identical, all behaving according to the same rules.

This multiplicity raises a foundational question: why are all electrons identical? The standard answer is that they are excitations of the same field, but this does not fully resolve the deeper issue of identity. The field itself is treated as a mathematical object, and the excitations are treated as separate entities. Nonlocality, coherence, and entanglement are then layered on top of this ontology, often requiring interpretational machinery such as wavefunction collapse, hidden variables, many worlds, or retrocausality.

The Single-Electron Field Identity (SEFI) framework takes a different approach. Instead of starting from many electrons, SEFI starts from one underlying entity. All observable electrons are manifestations of a single global mode of a geometric field. The electron is not a point particle but a stable curvature resonance, expressed repeatedly in spacetime. This shift in ontology has immediate consequences for identity, nonlocality, and the structure of quantum theory.

In this paper, I outline the core SEFI construction, focusing on the electron as a geometric resonance mode. I show how charge, mass, spin, and stability emerge from the same underlying structure, and how nonlocality becomes a natural consequence of unified identity. The goal is to provide a coherent geometric foundation that can sit beneath existing calculational frameworks, not to discard their predictive success.

## II. GEOMETRIC WARP-METRIC FRAMEWORK

### A. Curvature-Based Resonance Ontology

The SEFI framework is built on a warp-metric geometry in which curvature and torsion are allowed to vary in a controlled, mode-like fashion. Let the spacetime metric be written as

$$g_{\mu\nu}(x) = \eta_{\mu\nu} + h_{\mu\nu}(x),$$

where $\eta_{\mu\nu}$ is a reference background (e.g., Minkowski) and $h_{\mu\nu}(x)$ encodes warp and curvature associated with the electron mode. The key idea is that the electron is not a localized bump in a field, but a standing curvature pattern: a resonance of $h_{\mu\nu}(x)$ that is globally consistent and locally expressible.

We introduce a curvature-resonance field $\Phi(x)$ whose stable modes correspond to particle-like entities. In SEFI, the electron is identified with the first stable harmonic of $\Phi(x)$, subject to geometric constraints that enforce charge, mass, and spin properties. The ontology is thus:

> All electrons are expressions of the same global mode of $\Phi(x)$, realized as repeated local manifestations of a single curvature resonance.

### B. Mode Structure and Stability

Consider a resonance condition of the form

$$\Box\Phi(x) + V'(\Phi(x)) = 0,$$

where $\Box$ is the d'Alembertian associated with $g_{\mu\nu}(x)$ and $V(\Phi)$ is an effective potential that admits discrete stable modes. The electron mode $\Phi_e(x)$ is defined as the lowest nontrivial stable solution satisfying:

- **Charge constraint:** The mode carries a conserved current $J^\mu$ corresponding to charge $-e$.
- **Mass constraint:** The mode exhibits an effective inertial response consistent with $m_e$.
- **Spin constraint:** The mode transforms as a spin-1/2 representation under local Lorentz transformations.
- **Stability constraint:** The mode is dynamically stable under small perturbations.

These constraints are implemented geometrically by coupling $\Phi(x)$ to the warp-metric $g_{\mu\nu}(x)$ and to a spinor structure $\psi(x)$ that encodes the electron's spin behavior. The electron mode is thus a composite geometric object: a resonance of curvature, torsion, and spinor structure.

## III. SINGLE-ELECTRON IDENTITY

### A. From Many Electrons to One Entity

In standard quantum field theory, the electron field $\psi(x)$ admits many excitations, each corresponding to an electron. In SEFI, we reinterpret these excitations as local expressions of a single global mode. The field is not a container for many entities; it is the manifestation of one entity expressed in many locations.

Formally, we write the electron mode as

$$\psi_e(x) = \mathcal{M}[\Phi_e(x), g_{\mu\nu}(x)],$$

where $\mathcal{M}$ is a mapping from the curvature-resonance field and the warp-metric to a spinor field. The key ontological claim is:

There is one electron mode $\Phi_e(x)$, and all observed electrons are local manifestations of this mode.

This unified identity has immediate implications for nonlocality and coherence. If all electrons are the same entity, then correlated behavior is not mysterious; it is a consequence of the underlying unity.

### B. Nonlocality as a Consequence of Unified Identity

In SEFI, nonlocal correlations arise because the electron mode is globally defined. Local measurements probe different aspects of the same mode, not separate particles. The usual paradoxes of entanglement and nonlocality are reframed:

- There is no need for wavefunction collapse; measurement is a local interaction with a global mode.
- Hidden variables are unnecessary; the mode itself carries the structure of correlations.
- Many-worlds and retrocausality are not required; coherence is built into the unified identity.

Mathematically, we treat entangled states as structured configurations of the electron mode and associated fields, with correlations encoded in the geometry of $\Phi_e(x)$ and $g_{\mu\nu}(x)$. The global nature of the mode ensures that local operations can produce nonlocal effects without violating causality, because the underlying entity is already unified.

## IV. EMERGENT ELECTRON PROPERTIES FROM GEOMETRY

### A. Charge as Curvature Rotation

In SEFI, electric charge is modeled as a geometric rotation in a curvature-torsion space. The conserved current $J^\mu$ arises from a symmetry of the warp-metric and the resonance field:

$$J^\mu = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \theta)},$$

where $\theta$ is a phase-like geometric parameter associated with the electron mode. The value $-e$ is not an arbitrary assignment; it is a consequence of the mode's geometric structure.

### B. Mass as Mode Stiffness

Mass $m_e$ is interpreted as the stiffness of the electron mode against deformation. In the SEFI framework, the effective mass arises from the curvature energy of $\Phi_e(x)$:

$$m_e \sim \int d^3x \, \mathcal{E}_{\text{curv}}[\Phi_e(x), g_{\mu\nu}(x)],$$

where $\mathcal{E}_{\text{curv}}$ is an energy density functional. The electron's inertial behavior is thus tied directly to the geometry of its resonance.

### C. Spin as Geometric Orientation

Spin-1/2 behavior is encoded in the orientation of the electron mode within the local Lorentz structure. The spinor field $\psi_e(x)$ arises from the mapping $\mathcal{M}$ and inherits its transformation properties from the geometry of $\Phi_e(x)$ and $g_{\mu\nu}(x)$. Spin is not an abstract quantum number; it is a geometric orientation of the mode.

## V. DISCUSSION AND OUTLOOK

The SEFI framework proposes a unified geometric ontology for the electron. By treating all electrons as manifestations of a single global mode, SEFI resolves several foundational issues in quantum theory:

- The uniformity of electron properties becomes a geometric inevitability, not a coincidence.
- Nonlocality and coherence are natural consequences of unified identity.
- Charge, mass, spin, and stability emerge from the same curvature-resonance structure.

This paper has outlined the core construction and sketched the mathematical structure of the electron mode. A full treatment, including detailed warp-metric diagrams, explicit solutions for $\Phi_e(x)$, and applications to specific quantum phenomena, is presented in the accompanying monograph.

The intent is not to discard existing quantum field theory, but to provide a deeper geometric foundation that explains why its structures work and why electrons behave as they do. SEFI offers a way to unify ontology, geometry, and quantum behavior under one entity: the single-electron mode.

## ACKNOWLEDGMENTS

The author thanks the broader physics community for maintaining an open discourse on foundational questions, and acknowledges the independent nature of this work, developed outside traditional institutional frameworks.

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[1] Standard QFT reference here.
[2] Standard GR/geometry reference here.
[3] Foundations of quantum mechanics reference here.